Lab 07 – Non-parametric tests & Chi-squared tests

ENVX1002 Handbook

Semester 1, 2026

Learning Outcomes

- Learn to use R to perform a Wilcoxon signed-rank test and a Mann-Whitney U test.
- Learn to use R to calculate a chi-squared test for:
  - Test of goodness of fit
  - Test of independence
- Learn how to interpret statistical output.

Before we begin

Create your Quarto document and save it as `Lab-07.qmd` or similar. There are no data files to download for this lab.

Quick introduction to non-parametric tests

Non-parametric tests are statistical tests that do not assume a specific distribution for the data. They are often used when assumptions are uncertain or when a more robust, rank-based comparison is preferred. For example, non-parametric tests are useful when dealing with ordinal data or when the sample size is small.

The two non-parametric tests we will cover in this lab are:

1. Wilcoxon signed-rank test - used to compare one sample to a known value or to compare two related samples, matched samples, or repeated measurements on a single sample to assess whether their population mean ranks differ (similar to a one sample t-test or paired t-test).
2. Chi-squared test - used to compare observed frequencies to expected frequencies in categorical data. There are two types of chi-squared tests:

- Test of goodness of fit: tests whether the observed frequencies of a single categorical variable match expected frequencies based on a specific distribution.
- Test of independence: tests whether there is an association between two categorical variables.

Exercise 1: dog pain (walk-through)

Background

A study measures dogs' pain using the Glasgow CMPS-SF scale before and after medication. Note that this is ordinal data i.e. pain scale from 0 - 24 and the measurement is taken before and after medication - a great use case for the Wilcoxon Signed-Rank test

```r
before <- c(7, 6, 8, 5, 9, 7, 6, 8, 7, 5)
after  <-  c(4, 5, 6, 3, 7, 5, 4, 6, 5, 3)

## create a data frame
pain <- data.frame(
  dog_ID = 1:length(before),
  before = before,
  after = after,
  diff = after - before
)
```

EDA

```r
boxplot(pain$before, pain$after,
        names = c("Before", "After"),
        col = c("#00AFBB", "#E7B800"),
        ylab = "Pain Score",
      main = "Pain Scores Before and After Medication")
```

It looks like the median pain scores are lower after medication. But we need to test this statistically.

Hypothesis (H)

When in doubt write out the hypothesis in full:

- The null hypothesis is that there is no difference in pain scores before and after treatment.
- The alternative hypothesis is that there is a difference in pain scores before and after treatment.

Assumptions (A)

The key issue is that pain scores are ordinal (ranked), not continuous measurements. Therefore, a parametric test like the paired t-test is not strictly appropriate. The Wilcoxon signed-rank test is designed for paired ordinal or non-normal data.

The assumptions of the Wilcoxon signed-rank test are: 1. In this case the data are paired (i.e., the same subjects are measured before and after treatment). 2. the differences can be meaningfully ordered and are approximately symmetric about the median. Note that these tests are reasonably robust to some outliers.

So a histogram on the difference in scores for each pairs is good in this case.

```
ggplot(pain, aes(x = diff)) +
  geom_histogram(binwidth = 1)
```

We can see that the pain scores are reasonably symmetrical.

Test (T)

Note that just like a paired t-test we need to specify the data as below - this is in an effort to ensure the data ties are not lost!

```
# Perform the Wilcoxon signed-rank test
wilcox.test(x = pain$before, y = pain$after, paired
= TRUE)
```

The output shows the test statistic (V) and the p-value. The test statistic is the sum of the ranks of the positive differences (after - before). The p-value is the probability of observing a test statistic as extreme as the one obtained, assuming that the null hypothesis is true.

We also see that the test is Wilcoxon signed rank test with continuity correction. This is because the test is based on ranks, and the continuity correction is applied to adjust for the fact that we are using a discrete distribution to approximate a continuous distribution.

P-value (P)

We can see that the p-value is less than 0.05, so we reject the null hypothesis and conclude that there is a significant difference in pain scores before and after treatment.

Conclusion (C)

Given that the p-value is significant ($p < 0.05$) and looking at the output of the boxplot, we can conclude that the pain scores are significantly lower after treatment. This suggests the treatment reduced pain scores!

So can we use a Normal t-test?

Technically No (Strictly Speaking): The paired t-test requires interval or ratio data and normally distributed differences. Using it on ordinal data can yield inaccurate p-values and false conclusions because it treats ranked data (don't agree - neutral - agree) as if they were equally spaced measurements.

Exception: If the sample size is large (e.g., >30-40) and the ordinal data behaves close to a normal distribution, a t-test might provide similar conclusions, but the Wilcoxon test remains more robust for this data type.

Now analyse the data with a paired t-test. What do you find? Compare the p-values from the two tests.

- If the p-value from the t-test is smaller, this can suggest that the t-test is more powerful **when its assumptions are met**. The paired t-test assumes that the differences are approximately normally distributed and measured on a continuous scale.
- In this example, the data are ordinal and may not meet these assumptions, so the p-values from the t-test may not be reliable. The Wilcoxon signed-rank test is more appropriate here because it does not rely on normality and is suitable for ordinal data.
- When the assumptions of the t-test are violated, it may produce misleading results, whereas the Wilcoxon test provides a more robust alternative.

Exercise 2: Chi-squared tests

Quick introduction

The chi-square test is used to compare the *observed* distribution to an *expected* distribution, in a situation where we have **two or more categories** in discrete data. In other words, it compares observed counts in categories to the counts expected under a null hypothesis.

The formula is:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

where O_i is the observed frequency, E_i is the expected frequency for each category, and k is the number of categories.

For more information about the technique, consult your lecture slides and tutorial 7.

Wild tulips (walk-through)

Background

Suppose we collected wild tulips and found that 81 were red, 50 were yellow and 27 were white. Are these colours equally common?

The expected proportion would be 1/3 for each colour. Therefore, we want to test if the observed proportions are significantly different from the expected proportions.

The data is below.

```
tulip <- c(81, 50, 27)
```

Instructions

Utilise the **HATPC** process and test the hypothesis that the proportion of flower colours of tulips are equally common, assuming that the samples are independent. We can explore the data as we check the assumptions of the test.

HATPC:

- Hypothesis
- Assumptions
- Test (statistic)
- P-value
- Conclusion

> ℹ Level of significance
>
> **The level of significance is usually set at 0.05.** Using a smaller significance level makes it harder to reject the null hypothesis, which can reduce false positives but may increase the chance of missing a real effect.

Hypotheses

What are the null hypothesis and alternative hypotheses?

Click here to view answer

- H_0: The population proportions of red, yellow, and white tulips are equal, each 1/3.
- H_1: The population proportions of tulip colours are not all equal to 1/3.

Assumptions

Recall that the assumptions of the χ^2 test are:

1. No cell has expected frequencies less than 1
2. No more than 20% of cells have expected frequencies less than 5
3. The observations should also be independent.

> **Note**
>
> In the case that the above assumptions are violated then the probability of a type 1 error occurring (rejecting the null hypothesis when it is true, i.e. false positive) increases.

To calculate expected frequencies, we first calculate the total number of tulips and then divide by the number of categories.

```
expected <- rep(sum(tulip) * 1 / 3, 3) #rep function replicates the value we're calculating inside the brackets
expected
```

Does the data satisfy the assumptions of a χ^2 test?

Click here to view answer Yes, as expected frequencies > 5.

Test statistic

The `chisq.test()` function in R is used to calculate the chi-squared test.

```
res <- chisq.test(tulip, p = c(1 / 3, 1 / 3, 1 / 3))
res
```

Note that we could check our assumptions post-analysis by checking the expected frequencies stored in the `expected` object of the output:

```
res$expected
```

P-value

Write down how you should report the critical value, p-value and df in a scientific paper?

Click here to view answer $\chi^2 = 27.9, df = 2, p < 0.01$

Conclusions

Based on the p-value, do we retain or reject the null hypothesis?

Click here to view answer We reject the null hypothesis as the p-value is less than 0.05.

Now write a scientific (biological) conclusion based on the outcome.

Click here to view answer There is a significant difference in the proportion of flower colours of tulips ($\chi^2 = 27.9, df = 2, p < 0.01$).

Exercise 3: hermit crabs

Background

In a study of hermit crab behaviour at Point Lookout, North Stradbroke Island, a random sample of 3 types of gastropod shells was collected. Each shell was then scored as being either occupied by a hermit crab or empty. Do hermit crabs prefer a certain shell?

Shell species	Occupied	Empty
Austrochochlea	47	42
Bembicium	10	41
Cirithid	125	49

The data is stored in a `table` object in R below. Note that it is different from a `data.frame` object. You can verify this by using the `str()` or `class()` functions.

```
crabs <- as.table( #Make a table with values for each
row
  rbind(
    Aus = c(47, 42),
    Bem = c(10, 41),
    Cir = c(125, 49)
  )
)

colnames(crabs) <- c("Occupied", "Empty") #Add column
names to table
str(crabs)
crabs
```

Data exploration

Since we have a multi-dimensional dataset, we can try to plot the data to visualise it.

A mosaic plot is a graphical representation of the data in a two-way contingency table. It is a way of visualising the relationship between two categorical variables.

- width reflects the marginal totals for one variable
- height within each box reflects conditional proportions of the other variable

Try the code below. Can you interpret the plot?

```
# mosaic plot of crabs
mosaicplot(crabs, main = "Hermit crabs and shell species")
```

Click here to view interpretation The mosaic plot shows that shell species differ in their total frequencies and also in the proportion occupied versus empty. In particular, *Cirithid* shells appear to have a higher proportion occupied, while *Bembicium* shells have a relatively high proportion empty. This suggests occupancy may depend on shell species.

HATPC Analysis

Now it is your turn to test the hypothesis that the three shell species are equally preferred by hermit crabs. Follow the HATPC process with the following questions in mind (but you don't have to answer them individually):

1. What are the null hypothesis and alternative hypotheses?
2. Does the data satisfy the assumptions of a χ^2 test?
3. How should you report the critical value, p-value and df in a scientific paper?
4. Based on the p-value, do we retain or reject the null hypothesis?
5. Write a scientific (biological) conclusion based on the outcome.

Take your time, and when you are ready, check your answers with your demonstrators.

⚠️ Answer to Exercise 3

Hypothesis

H_0: Occupancy status (occupied or empty) is independent of shell species.

H_1: Occupancy status depends on shell species.

Assumptions

To check whether the expected frequencies are greater than 5, we can either perform the calculation manually or check the output of the `chisq.test()` function.

```
fit <- chisq.test(crabs)
fit$expected
```

From the output, we can see that all expected frequencies are greater than 5. The other assumption where no cell has expected frequencies less than 1 is also satisfied.

Test statistic

If you have not already done so, perform the chi-squared test using the `chisq.test()` function in R.

```
fit <- chisq.test(crabs)
fit
```

We reject the null hypothesis as the p-value is less than 0.05 and conclude that there is strong evidence that shell occupancy is associated with shell species. ($\chi^2 = 45.5, df = 2, p < 0.01$).

Exercise 4: pregnancies

Quick introduction

The Mann-Whitney U test (also called the Wilcoxon rank-sum test) is a non-parametric alternative to the independent samples t-test. It's used to compare two independent groups when the data doesn't meet the assumptions for a t-test, particularly when:

- The data is not normally distributed
- The sample sizes are small
- The data is ordinal rather than continuous

The test works by ranking all observations from both groups together, then comparing the sum of ranks between the two groups. If there's a significant difference in these rank sums, it suggests that the distributions of the two groups differ.

Background

In a medical study, researchers measured the permeability constants of the human chorioamnion (a placental membrane) between different gestational ages. They collected samples from:

- Term pregnancies (x): pregnancies at full term (around 40 weeks)
- Early pregnancies (y): pregnancies between 12 to 26 weeks gestational age

The researchers wanted to test if the permeability of the membrane is greater at term compared to earlier in pregnancy. Higher permeability might indicate changes in the membrane's function as pregnancy progresses.

```
# Permeability constants at term (x)
term <- c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91,
1.64, 0.73, 1.46)

# Permeability constants at 12-26 weeks (y)
early <- c(1.15, 0.88, 0.90, 0.74, 1.21)

# Create a data frame for visualisation
placenta <- data.frame(
  permeability = c(term, early),
     group   =   factor(c(rep("Term",   length(term)),
rep("Early", length(early))))
)
```

EDA

Let's visualise the data using boxplots to see if there are visible differences between the two groups:

```
# Create a comparative boxplot
ggplot(placenta, aes(x = group, y = permeability, fill
= group)) +
  geom_boxplot() +
  geom_jitter(width = 0.2, alpha = 0.6) +
   labs(title = "Placental Membrane Permeability by
Gestational Age",
```

```
       x = "Pregnancy Stage",
       y = "Permeability Constant") +
  theme_minimal()
```

Also, let's check if the data is normally distributed using QQ plots:

```
# QQ plots for both groups
par(mfrow = c(1, 2))
qqnorm(term, main = "QQ Plot for Term Pregnancies")
qqline(term)
qqnorm(early, main = "QQ Plot for Early Pregnancies")
qqline(early)
```

Hypothesis (H)

Based on the research question, we can formulate the following hypotheses:

- $\mathbf{H_0}$: The distribution of permeability at term is the same as or lower than early pregnancy.
- $\mathbf{H_1}$: The distribution of permeability at term is greater than early pregnancy.

Note that this is a one-sided test, as we're specifically interested in whether the permeability is *greater* at term, not just whether it differs.

Assumptions (A)

The Mann-Whitney U test has the following assumptions:

1. The observations from both groups are independent of each other.
2. The observations are ordinal (i.e., can be ranked).
3. The distributions of both populations have the same shape (though this is relaxed for large samples).

From our EDA, we can see that:

- Given the small sample sizes and potential deviations from normality, a rank-based test such as the Mann–Whitney U test provides a more robust comparison than a t-test.
- The observations are certainly independent as they come from different individuals.

Given these observations, the Mann-Whitney U test is an appropriate choice over a t-test. Note that we may run a formal test for normality (e.g., Shapiro-Wilk test) if we wanted to be thorough.

With such small sample sizes, QQ plots are difficult to interpret, so we rely more on robust methods. If the assumptions of the t-test are satisfied, it is generally preferred because it is more powerful. The Mann–Whitney test remains a valid alternative but may be less efficient than a t-test (i.e., it may have a higher chance of failing to detect a real effect - https://en.wikipedia.org/wiki/Power_(statistics)).

Test (T)

We'll use the `wilcox.test()` function to perform the Mann-Whitney U test. Since we're interested in whether the permeability is greater at term, we'll use a one-sided alternative hypothesis:

```
# Perform the Mann-Whitney U test
wilcox.test(term, early, alternative = "greater")
```

We can also use the large-sample approximation method (without continuity correction) as described by Hollander & Wolfe:

```
# Mann-Whitney U test with large sample approximation
wilcox.test(term, early, alternative = "greater",
            exact = FALSE, correct = FALSE)
```

P-value (P)

The p-value from the test is 0.1914, which is greater than our significance level of 0.05.

Conclusion (C)

Based on the results of the Mann-Whitney U test, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the permeability of the human chorioamnion is greater at term compared to earlier in pregnancy (12-26 weeks).

Despite what we might observe visually in the boxplots, the statistical test doesn't support the claim of higher permeability at term. This could be due to several factors:

1. Small sample sizes (only 10 term and 5 early samples)
2. High variability within the term group
3. The actual biological relationship might be more complex than our simple hypothesis

Question 1

Why might researchers choose a Mann-Whitney U test over a t-test in this situation? Consider both the sample size and what you observed in the QQ plots.

Question 2

What does the value "W = 31" in the test output represent? What information does it give us about the comparison between the two groups?

⚠️ Answer to Exercise 4

Question 1

Researchers might choose the Mann-Whitney U test over a t-test for several reasons:

1. The sample sizes are small (10 and 5), which makes it difficult to reliably assess normality.
2. The QQ plots show potential deviations from normality, especially for the term pregnancy group.
3. The Mann-Whitney U test doesn't assume normality, making it more robust for small or non-normal samples.
4. When dealing with biological measurements that might have outliers or skewed distributions, non-parametric tests often provide more reliable results.

Question 2

The value "W = 31" represents the sum of ranks for the first group (term) after all observations from both groups are ranked together.

In the Mann-Whitney U test:

- All observations from both groups are combined and ranked from lowest to highest
- The test statistic W is the sum of ranks assigned to the first group
- If the first group tends to have higher values, W will be larger than expected under the null hypothesis
- If the first group tends to have lower values, W will be smaller than expected

In this case, W = 31 wasn't extreme enough to reject the null hypothesis, suggesting that the distribution of permeability values isn't significantly higher in term pregnancies compared to early pregnancies.

Bonus take home exercises

Exercise 1: National RSPCA statistics

The RSPCA releases statistics on the number of animals they receive, reclaim and rehome every year. In the 2023-24 financial year, the RSPCA received 17468 dogs, 26704 cats, and 37497 other

animals. The "other" category includes horses, small animals, livestock and wildlife.

Using the HATPC framework, test whether these animals were received in equal proportions.

```
received <- c(17468, 26704, 37497)
```

⚠️ Solution to Exercise 1: National RSPCA statistics

Hypothesis

H_0 : There is no significant difference between the observed and the expected number of animals received from each category

H_1 : There is a significant difference between the observed and the expected number of animals received from each category

Assumptions

```
expected_received <- rep(sum(received)*1/3, 3)
expected_received
```

1) No cell has expected frequencies less than 1 and 2) No more than 20% of cells have expected frequencies less than 5 so the assumptions are met

Test (statistic)

```
chisq.test(received)
```

P-value

$p < 0.05$, so we reject the null hypothesis

Conclusion

There is a significant difference between the observed and the expected number of animals received from each category

Exercise 2: UC Berkeley Admissions

For the two exercises we're going to use simplified versions of the inbuilt dataset 'UCBAdmissions', which has data on student admissions for Berkeley. The dataset shows how many students were rejected and admitted to the university by both department and gender.

2.1 Admissions by department

Did every department at UC Berkeley admit students in equal proportions?

```
dept_admissions <- c(601, 370, 322, 269, 147, 46)
```

⚠️ Solution to Exercise 2.1: Admissions by department

Hypothesis

H_0: There is no significant difference between the expected and observed number of students admitted by each department

H_1: There is a significant difference between the expected and observed number of students admitted by each department

Assumptions

```
rep(sum(dept_admissions)*1/6, 6)
```

1) No cell has expected frequencies less than 1 and 2) No more than 20% of cells have expected frequencies less than 5 so the assumptions are met

Test (statistic)

```
chisq.test(dept_admissions)
```

P-value

$p < 0.05$, so we reject the null hypothesis

Conclusion

There is a significant difference between the expected and observed number of students admitted by each department

2.2 Gender differences in admissions

Are male and female students admitted and rejected in the same proportion?

```
gender <- as.table( #Make a table with values for each
row
  rbind(
    admitted = c(1198, 557),
    rejected = c(1493, 1278)
```

```
  )
)

colnames(gender) <- c("Male", "Female") #Add column
names to table
```

⚠️ Solution to Exercise 2.2: Gender differences in admissions

Hypothesis

H_0: There is no significant difference between the expected and observed number of students admitted or rejected by gender

H_1: There is a significant difference between the expected and observed number of students admitted or rejected by gender

Assumptions

```
# Fit the chisq test to be able to call up the
expected
fit_gender <- chisq.test(gender)
fit_gender$expected
```

1) No cell has expected frequencies less than 1 and 2) No more than 20% of cells have expected frequencies less than 5 so the assumptions are met

Test (statistic)

```
# Call up the test we ran above
fit_gender
```

P-value

$p < 0.05$, so we reject the null hypothesis

Conclusion

There is a significant difference between the expected and observed number of students admitted or rejected by gender